

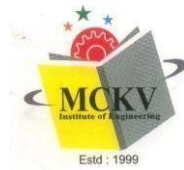
AgroCare: Plant disease detection & Crop Recommendation System

Submitted as a partial fulfillment of Bachelor of Technology in Computer Science & Engineering
Of

MCKV Institute of Engineering
(An Autonomous Institute under UGC Act, 1956)

Approved by AICTE

Affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal)



Project Term Paper

AgroCare: Plant disease detection & Crop Recommendation System

Submitted by
Project Group Number: 4

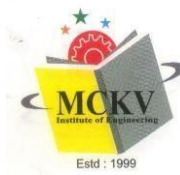
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CERTIFICATE OF RECOMMENDATION

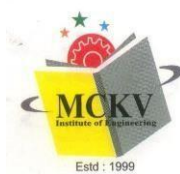
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for the project entitled **“AgroCare: Plant disease detection & Crop Recommendation System”** to be accepted in partial fulfillment of the requirements for the degree of **Bachelor of Technology in Computer Science & Engineering**.

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CERTIFICATE OF APPROVAL

(B.Tech Degree in Computer Science & Engineering)

This project term paper is hereby approved as a creditable study of an engineering subject carried out and presented in a satisfactory way to warrant its acceptance as a prerequisite to the degree for which it has been submitted. It is to be understood that by this approval, the undersigned does not necessarily endorse or approve any statement made, opinion expressed, and conclusion drawn therein but approves the project term paper only for the purpose for which it has been submitted.

COMMITTEE ON FINAL
EXAMINATION FOR
EVALUATION OF
PROJECT TERM PAPER

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ABSTRACT

Agriculture is fundamental to ensuring food security and economic stability, especially in regions reliant on farming. AgroCare is an innovative AI-driven system designed to assist farmers by integrating Plant Disease Detection and Crop Recommendation functionalities. Leveraging deep learning, machine learning, and IoT technologies, AgroCare empowers farmers with real-time insights to enhance productivity and promote sustainable farming practices.

The Plant Disease Detection module employs a Convolutional Neural Network (CNN) developed with PyTorch, trained on the Plant Village dataset, to identify 39 plant diseases from leaf images. Integrated with a user-friendly Flask web application and supported by edge computing devices like Raspberry Pi and NVIDIA Jetson Nano, it provides on-device disease identification and treatment suggestions.

The Crop Recommendation System utilizes machine learning models to analyze soil (pH, nitrogen, phosphorus, potassium) and environmental parameters (temperature, humidity, rainfall) for recommending suitable crops. Based on extensive studies and inspired by recent advancements in crop yield prediction, the system ensures data-driven and region-specific crop suggestions. IoT-based soil and weather sensors, combined with Raspberry Pi/Arduino controllers, enable real-time monitoring for precision agriculture.

To ensure inclusivity, AgroCare also offers external devices for low-income farmers, eliminating dependence on smartphones. Open-source by design, AgroCare encourages community contributions to enhance its performance and accessibility, aiming to revolutionize farming with smart, affordable technology.

INTRODUCTION

1. Overview

Agriculture plays a pivotal role in sustaining economies, providing food security, employment, and contributing to national income, especially in developing countries. Despite advancements in farming methods, farmers continue to face several critical challenges, including plant diseases, inefficient crop management, soil degradation, and the unpredictable impacts of climate change. Addressing these challenges requires a shift from traditional practices to modern, technology-driven solutions.

Recent developments in Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and the Internet of Things (IoT) have opened new avenues for transforming agriculture. Predictive analytics, real-time monitoring, and data-driven recommendations are now increasingly accessible to farmers, thanks to advances in computational power and sensor technology. These technologies enable early detection of plant diseases, optimization of crop choices, efficient resource management, and overall improvement in agricultural productivity.

AgroCare is a proposed AI-powered system that integrates Plant Disease Detection and Crop Recommendation into a unified platform. By using Convolutional Neural Networks (CNNs) for image-based disease diagnosis and machine learning algorithms for crop recommendations based on soil and environmental parameters, AgroCare empowers farmers with actionable insights in real time. The system leverages edge computing devices like Raspberry Pi and NVIDIA Jetson Nano along with IoT-based sensors to enable continuous monitoring and on-site decision-making without heavy cloud dependency.

2. Project Motivation

The motivation behind AgroCare stems from the pressing need to modernize agricultural practices, particularly in resource-constrained regions. Several key factors inspired the development of this project:

- **High Crop Losses Due to Disease:** Farmers often incur significant financial losses due to late detection and misdiagnosis of plant diseases. Early and accurate identification is crucial to prevent widespread crop damage.
- **Ineffective Crop Selection:** Choosing the wrong crop for a particular soil type or climatic condition leads to low yield and inefficient resource utilization. A system that recommends the most suitable crops based on data analysis can significantly improve outcomes.

- **Lack of Technological Accessibility:** While smartphone applications for smart farming exist, many small and marginal farmers lack access to smartphones and high-speed internet. AgroCare is designed with accessibility in mind, using affordable external devices to ensure inclusivity.
- **Growing Research Interest:** Studies have consistently demonstrated the effectiveness of machine learning and deep learning models in predicting crop yields and diseases. Inspired by this body of research, AgroCare integrates these advanced methodologies into a practical, deployable solution.
- **Climate Change and Sustainability Concerns:** The variability introduced by climate change necessitates smarter and more adaptive farming techniques. AgroCare supports sustainable agriculture by helping farmers make data-backed decisions that optimize resource usage and reduce environmental impact.

3. Significance of the Study

The significance of AgroCare lies in its potential to revolutionize farming practices through the integration of advanced technologies. Some of the key contributions and impacts of this project are:

- **Empowering Farmers with Knowledge:** AgroCare provides farmers with immediate, actionable information about plant health and crop selection, helping them make informed decisions and increasing their confidence in managing their fields.
- **Enhancing Agricultural Productivity:** By enabling early disease detection and recommending crops suited to local soil and environmental conditions, AgroCare improves crop yield, quality, and profitability.
- **Promoting Sustainable Farming:** Precision agriculture techniques enabled by AgroCare lead to optimized use of water, fertilizers, and pesticides, contributing to environmental conservation and sustainable resource management.
- **Bridging the Digital Divide:** The project addresses the technological gap in rural areas by offering an external device solution for farmers without smartphones, ensuring that innovation is inclusive and accessible.
- **Supporting Open-Source Development:** As an open-source initiative, AgroCare encourages community collaboration, allowing researchers and developers worldwide to contribute toward enhancing its functionality, scalability, and user experience.
- **Alignment with Global Agricultural Goals:** AgroCare's objectives align with international efforts such as the UN Sustainable Development Goals (SDGs), particularly Goal 2: Zero Hunger, by promoting sustainable agriculture and ending food insecurity.

SCOPE OF THE WORK

The **AgroCare** project focuses on designing and developing an integrated system that combines **Plant Disease Detection** and **Crop Recommendation** to empower farmers with real-time, AI-driven agricultural insights. The system aims to leverage recent advancements in Deep Learning, Machine Learning, and IoT to address persistent challenges in farming such as delayed disease detection, poor crop selection, and inefficient resource utilization. By employing technologies like Convolutional Neural Networks (CNNs), edge computing devices (e.g., Raspberry Pi, NVIDIA Jetson Nano), and IoT-based environmental sensors, AgroCare ensures quick, accurate, and localized decision support for farmers, even in regions with limited internet connectivity.

The scope extends to building a web-based platform where farmers can upload images of diseased plants for instant analysis, as well as a real-time monitoring system that captures soil and environmental parameters to suggest the most suitable crops. Furthermore, the system is designed with inclusivity in mind, offering external devices for farmers who do not have smartphones. AgroCare also embraces open-source development, allowing for future expansions such as the inclusion of more plant species, crops, regional datasets, and advanced AI models.

The specific deliverables for the two major components are as follows:

Plant Disease Detection System:

- A deep learning model (CNN) trained to identify 39 plant diseases using leaf images.
- A Flask-based web application that enables farmers to upload images and receive disease classification and treatment recommendations.
- Integration with Raspberry Pi or NVIDIA Jetson Nano for real-time, on-device inference without constant internet dependency.
- Support for capturing leaf images using camera modules connected to IoT devices.

Crop Recommendation System:

- An IoT-based data collection framework for capturing soil pH, nitrogen, phosphorus, potassium levels, temperature, humidity, and rainfall.
- A machine learning model trained to recommend the most suitable crops based on real-time soil and environmental data.
- Real-time data processing and analysis using edge computing platforms.
- An accessible external device designed for farmers lacking smartphones, delivering crop recommendations in a simplified format.

By focusing on these deliverables, AgroCare aims to significantly enhance agricultural productivity, promote sustainable resource management, and bridge the technological gap faced by many small and marginal farmers.

STUDY OF PREVIOUS WORK

Agricultural practices have evolved significantly over the decades, moving from traditional observational methods to data-driven and AI-supported solutions. A review of past research, agricultural guidelines, and modern technical practices provides valuable insights into the development of systems like AgroCare.

Traditional Approaches in Agriculture

In early agricultural practices, soil fertility and crop management were largely based on empirical knowledge passed down through generations. Texts such as *Soil Fertility and Fertilizers* by *Tisdale et al. (1999)* highlight the importance of understanding nutrient cycles and soil amendments to optimize crop yields. Farmers historically relied on physical symptoms such as leaf discoloration, wilting, and deformities for plant disease identification. However, these techniques were subjective and heavily dependent on a farmer's experience.

The Food and Agriculture Organization (FAO) (2021) provided formalized guidelines for soil and plant analysis, establishing standardized practices for soil testing, nutrient management, and crop recommendations. Despite such advancements, traditional methods remained limited in scalability and precision, often leading to inconsistent results.

Evolution towards Machine Learning and Deep Learning

The last decade witnessed a significant transition towards using Machine Learning (ML) and Deep Learning (DL) to solve complex agricultural problems:

Crop Yield Prediction:

Studies such as *Nikhil et al. (2024)* and *Bogireddy & Murari (2024)* demonstrated the use of Random Forest classifiers and boosting techniques to predict crop yields based on soil and climatic data. These methods significantly outperformed earlier statistical models in terms of accuracy.

Plant Disease Detection:

Deep learning approaches, particularly Convolutional Neural Networks (CNNs), were successfully applied in the works of *Jha et al. (2024)* and *Kaushik et al. (2024)* for diagnosing diseases from leaf images. Transfer learning and ensemble models further enhanced the robustness of predictions.

Climate-Smart Agriculture:

Research by *Hu et al. (2024)* highlighted how machine learning models, when combined with climate change projections, could improve resilience by suggesting better crop varieties for changing conditions.

While these studies greatly improved the precision of agricultural predictions, they typically focused on single problems (either disease detection or yield prediction) and often depended on cloud-based architectures, limiting applicability in rural areas with poor connectivity.

Best Practices in System Integration and Deployment:

From a technological perspective, robust and scalable agricultural systems benefit from sound frontend-backend integration and deployment practices. Guidelines from resources like [MDN Web Docs](#) and [Stack Overflow](#) discussions outline important principles for ensuring smooth communication between user interfaces and backend servers, essential for maintaining responsive applications even under limited bandwidth conditions.

Deployment techniques such as those detailed in [AWS Documentation on Elastic Beanstalk](#) offer practical insights into scalable, cloud-ready application hosting. However, edge computing has emerged as a complementary solution where cloud reliance is minimized—an approach AgroCare embraces through local inference using Raspberry Pi and Jetson Nano devices.

Recent Agricultural Extension Practices:

Extension programs, like those from the *University of Minnesota Extension*, continue to emphasize best planting and care practices for crops (e.g., apple trees), offering location-specific guidelines to farmers. These serve as foundational models for AgroCare's philosophy of region-specific crop recommendations, enhanced through data-driven decision support systems.

Distinction of AgroCare:

Building on these previous works, AgroCare distinguishes itself by integrating plant disease detection and crop recommendation into a single, edge-computing-supported platform. It is designed for offline operation, ensuring usability in areas with minimal internet connectivity. Additionally, AgroCare specifically addresses the technology gap by offering affordable external devices for farmers without smartphones, making modern agricultural technology accessible to all.

Unlike traditional isolated systems or purely cloud-dependent architectures, AgroCare leverages IoT-based real-time data collection, CNN-based disease detection, and ML-driven crop recommendation, thereby providing farmers with a complete, practical, and inclusive smart farming solution.

HARDWARE AND SOFTWARE DETAILS

i. Hardware

1. Name of the Hardware: Soil Sensors

Purpose of the Hardware: Real-time data collection from soil to monitor parameters like moisture, pH, and temperature for agricultural analysis **Specification:** Multi-parameter support (moisture, temperature, EC, pH); digital output with RS485/analog 0-5V; waterproof IP68 casing

Model/ Built: Dragino LSE01 / Vegetronix VH400 / DFROBOT SEN0308

Price (Approx.): ₹3,000 – ₹12,000 (~\$40 – \$150) depending on sensor type and brand

2. Name of the Hardware: Cameras

Purpose of the Hardware: Image data collection for computer vision-based plant health monitoring or yield estimation

Specification: 1080p/4K resolution, night vision optional, USB/HDMI interface, compatible with computer vision libraries

Model/ Built: Logitech C920 / Raspberry Pi Camera Module v2 / Intel RealSense D435

Price (Approx.): ₹4,000 – ₹25,000 (~\$50 – \$300) depending on model and features

ii. Software

1. Name of the Package/ Tool: Python (v3.8 or later)

Purpose of use: Primary programming language for data analysis, model training, and scripting

Licensed/ Free/ Open Sourced: Open Source <https://www.python.org/downloads/>

2. Name of the Package/ Tool: Jupyter Notebook / VS Code / PyCharm **Purpose of use:** Integrated Development Environment (IDE) for coding and visualizing results interactively

Licensed/ Free/ Open Sourced:

a. Jupyter Notebook: Open Source <https://jupyter.org/>

b. VS Code: Free (Microsoft) <https://code.visualstudio.com/>

- c. PyCharm: Community Edition - Free; Professional Edition - Licensed
<https://www.jetbrains.com/pycharm/>

3. **Name of the Package/ Tool:** Scikit-learn (v1.0+)

Purpose of use: Machine learning algorithms for data preprocessing and modeling

Licensed/ Free / Open Sourced: Open Source <https://scikit-learn.org/>

4. **Name of the Package/ Tool:** TensorFlow (v2.6+) or PyTorch (v1.10+)

Purpose of use: Deep learning model development and training

Licensed/ Free / Open Sourced:

- a. TensorFlow: Open Source <https://www.tensorflow.org/>
- b. PyTorch: Open Source <https://pytorch.org/>

5. **Name of the Package/ Tool:** Pandas (v1.3+)

Purpose of use: Data manipulation and analysis

Licensed/ Free / Open Sourced: Open Source <https://pandas.pydata.org/>

6. **Name of the Package/ Tool:** NumPy (v1.21+)

Purpose of use: Numerical computations and matrix operations

Licensed/ Free / Open Sourced: Open Source <https://numpy.org/>

7. **Name of the Package/ Tool:** Matplotlib C Seaborn

Purpose of use: Data visualization and plotting

Licensed/ Free / Open Sourced:

- a. Matplotlib: Open Source <https://matplotlib.org/>
- b. Seaborn: Open Source <https://seaborn.pydata.org/>

8. **Name of the Package/ Tool:** MySQL 8+ / PostgreSQL 13+ / Firebase **Purpose of use:** Database management for structured and real-time data storage

Licensed/ Free / Open Sourced:

- a. MySQL: Open Source <https://www.mysql.com/>
- b. PostgreSQL: Open Source <https://www.postgresql.org/>
- c. Firebase: Freemium (Google) <https://firebase.google.com/>

9. **Name of the Package/ Tool:** Flask/ Django (for web), React Native/ Flutter (for mobile)

Purpose of use: Backend & frontend development for deploying applications

Licensed/ Free/ Open Sourced:

- a. Flask: Open Source <https://flask.palletsprojects.com/>
- b. Django: Open Source <https://www.djangoproject.com/>
- c. React Native: Open Source <https://reactnative.dev/>
- d. Flutter: Open Source <https://flutter.dev/>

DETAILS ABOUT THE DATASETS

1. PLANT DISEASE DETECTION SYSTEM

Objective

The main aim of this part of the project is to identify plant diseases early through image-based diagnosis. This helps farmers take timely action to mitigate crop loss. The system leverages deep learning, specifically **Convolutional Neural Networks (CNNs)**, to classify leaf images into 39 different categories, each corresponding to a specific plant disease or a healthy plant condition.

Dataset Overview

Dataset Source:

Mendeley Data Repository – “Identification of Plant Leaf Diseases Using a 9-layer Deep Convolutional Neural Network”

- **Published On:** 18 April 2019
- **DOI:** 10.17632/tywbtsjrjv.1
- **Contributors:** Arun Pandian J, Geetharamani Gopal
- **Link to Dataset:** <https://data.mendeley.com/datasets/tywbtsjrjv/1>

Dataset Description

This dataset is extensively used for plant disease identification research and contains **61,486 images** of plant leaves categorized into **39 different classes**. These classes represent both **infected** and **healthy** leaves across a wide variety of crops such as Apple, Corn, Tomato, Potato, Grape, and more. It also includes **background images without leaves** for robustness.

Each image is high quality and labeled, making it suitable for training deep learning models.

Data Augmentation Techniques Used

To improve model generalization and enhance dataset diversity, six augmentation techniques were applied:

- **Image Flipping:** Horizontally or vertically flipping the images.
- **Gamma Correction:** Adjusts the brightness.
- **Noise Injection:** Adds random noise to simulate real-world conditions.
- **PCA Color Augmentation:** Alters the color channels for robustness.
- **Rotation:** Random rotations of images.
- **Scaling:** Resizing to different scales.

Dataset Files

- Plant_leaf_diseases_dataset_with_augmentation.zip (905 MB)
- Plant_leaf_diseases_dataset_without_augmentation.zip (828 MB)

Class Labels (39 Total)

Some of the key class labels include:

- Apple_scab, Apple_black_rot, Apple_cedar_apple_rust, Apple_healthy
- Corn_gray_leaf_spot, Corn_common_rust, Corn_northern_leaf_blight, Corn_healthy
- Tomato_early_blight, Tomato_late_blight, Tomato_yellow_leaf_curl_virus, Tomato_healthy
- Grape_black_rot, Grape_leaf_blight, Grape_healthy

...and many more.

Model Training Info

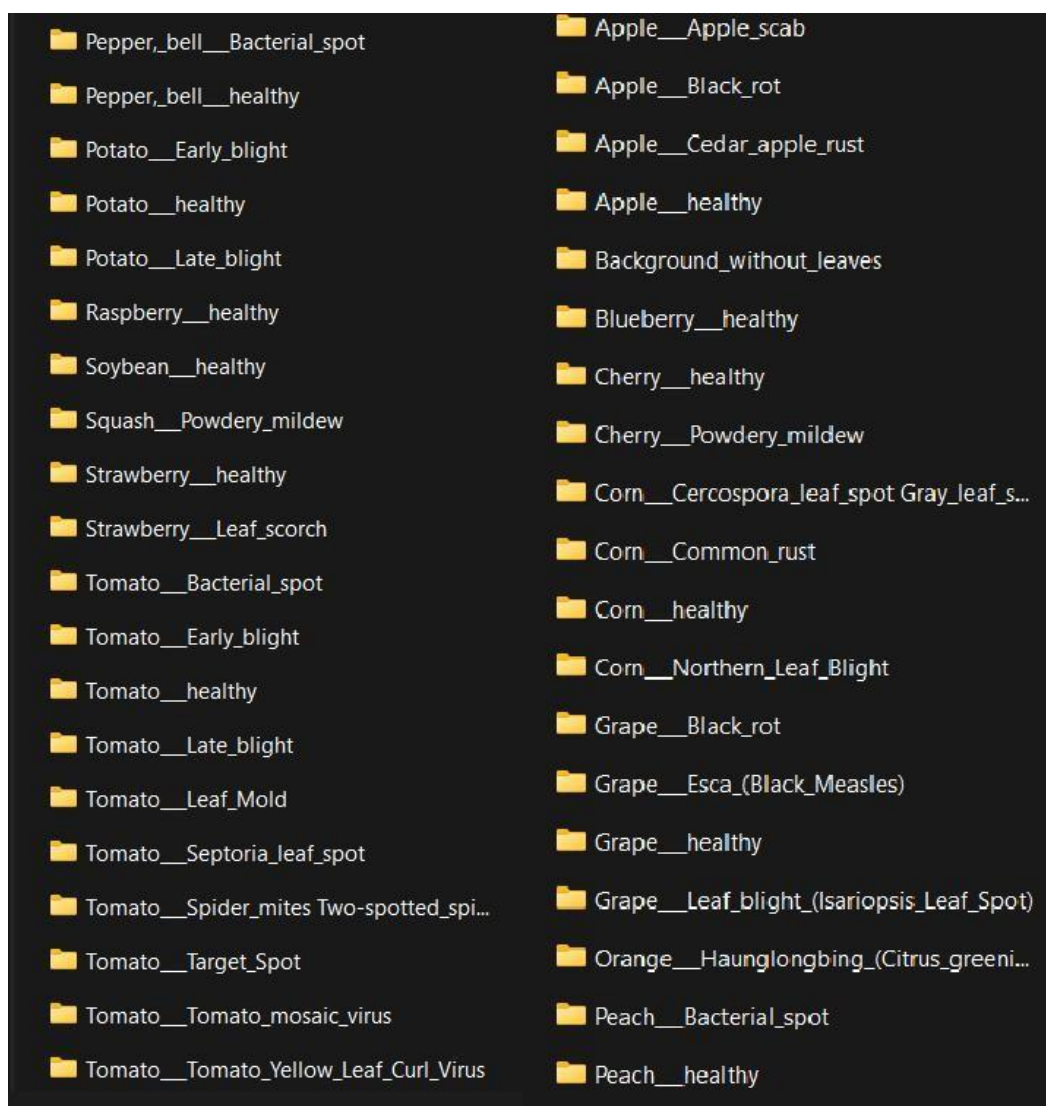
- **Framework:** PyTorch

- **Model:** CNN-based architecture (9-layer Deep CNN)
- **Pre-trained Model File:** plant_disease_model_1.pt
- **Deployment:** Web interface using Flask for real-time predictions.

Supplementary Files Used

1. **disease_info.csv:** Contains disease names, their symptoms, and additional descriptions to provide users with context on model predictions.
2. **supplement_info.csv:** Maps each disease to recommended supplements or treatments and includes direct product links for ease of purchase.

The view of the data set:



2. CROP RECOMMENDATION SYSTEM

Objective

The second part of the project focuses on helping farmers make **data-driven decisions** about which crops to plant based on environmental and soil conditions. This predictive model increases the efficiency of land usage and optimizes crop yields.

Dataset Overview

Dataset Source: Kaggle - Crop Recommendation Dataset

Dataset Description

This dataset is a part of **precision agriculture initiatives**, combining several data sources such as **climate, soil nutrients, and rainfall** to provide insights on the most suitable crops for cultivation in a given region.

The dataset was created by augmenting existing open datasets specific to **Indian agriculture**, and it can be used to train models that recommend crops based on measured inputs.

Key Variables

Variable	Description
N	Nitrogen content ratio in the soil
P	Phosphorus content ratio in the soil
K	Potassium content ratio in the soil
Temperature	Average temperature in degrees Celsius

Variable	Description
Humidity	Relative humidity (%)
pH	Soil pH level
Rainfall	Average annual rainfall (in millimeters)

Use Case Scenario

When a farmer inputs the values of N, P, K, temperature, humidity, pH, and rainfall from their local region, the model will process this information and recommend **the most suitable crop** for their land. This helps optimize yields and ensures effective use of fertilizers and water.

Model Info

- **Model Type:** Supervised classification model
- **Possible Models:** Decision Trees, Random Forests, KNN, or XGBoost
- **Input:** Numerical values of 7 features
- **Output:** One of many crop labels (e.g., rice, wheat, maize, etc.)

Conclusion

The **AgroCare** project successfully bridges deep learning and machine learning technologies with real-world agriculture. By integrating a **Plant Disease Detection System** using CNNs and a **Crop Recommendation System** trained on essential soil and climate parameters, it provides a dual solution to key farming challenges.

This system not only helps farmers detect plant diseases early through image analysis but also guides them in selecting the most suitable crops to grow based on their local environmental data. With its user-friendly interface and practical outputs, AgroCare can significantly enhance agricultural productivity and sustainability.

In essence, AgroCare is a smart farming assistant built for modern agriculture—designed to increase yields, reduce crop losses, and empower farmers through technology.

ROADMAP OF THE FUTURE WORK

Proposed Work for the Seventh Semester

In the seventh semester, the focus will be on enhancing the Plant Disease Detection model by expanding plant categories, optimizing model architecture with advanced CNNs like ResNet and EfficientNet, and applying transfer learning for improved accuracy. Simultaneously, the Crop Recommendation System will integrate IoT-based real-time soil and environmental sensing using devices like Raspberry Pi or Arduino, ensuring dynamic and accurate recommendations. Hardware and software integration will be achieved through real-time image capture and local inference via Jetson Nano for disease detection, and an embedded sensor network for crop recommendation. To promote accessibility, a low-cost external device supporting regional languages and simple UX will be developed for farmers without smartphones. Pilot deployments in farms will be conducted to gather feedback for iterative improvement, and the project will be opened to the global community via GitHub to encourage collaborative enhancements.

Proposed Work for the Eighth Semester

In the eighth semester, efforts will shift towards full-scale deployment, integrating the backend with a responsive frontend to deliver a user-friendly, cross-platform application hosted on cloud platforms like AWS or GCP. Scalable APIs, real-time data visualization, CI/CD pipelines, and secure user authentication will be implemented to ensure a production-ready solution. The AgroCare system will be cloud-deployed, with synchronization between edge devices and cloud servers to enable real-time updates. Advanced techniques like AutoML and Federated Learning will be explored for model improvement and privacy-preserving training. Comprehensive dashboards for farmers and admins will be developed, offering crop health tracking, analytics, and data export options. Additionally, the team will aim to publish a research paper, participate in hackathons, and finalize key deliverables including a mobile app, IoT kits, satellite monitoring module, and a sustainability plan to transform AgroCare into a long-term impactful platform for smart agriculture.

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